

Computer Science – Python Programming

11 January 2026 20:43

Time: 90 Minutes

Maximum Marks: 70

Instructions:

1. All questions are compulsory.
2. Read the questions carefully.
3. Assume suitable values wherever required.
4. Use proper indentation while writing programs.

Section A – Fill in the Blanks

(10 × 1 = 10 Marks)

1. The keyword used to define a function in Python is _____.
2. The operator used for floor division is _____.
3. A list is created using _____ brackets.
4. The function used to get input from the user is _____.
5. The statement used to handle exceptions is _____ block.
6. The data type of (10, 20, 30) is _____.
7. The operator ** is used for _____ operation.
8. In Python, dictionary stores data in _____ pairs.
9. The keyword used to exit a loop is _____.
10. len() function is used to find the _____ of a sequence.

Section B - True / False

(10 × 1 = 10 Marks)

1. Python is a case-sensitive language.
2. Tuples are mutable data types.
3. break statement terminates the loop completely.
4. match..case is used for multiple condition checking.
5. // gives the exact division result.
6. Strings in Python can be modified.
7. A dictionary allows duplicate keys.
8. try block is used to handle runtime errors.
9. and is a logical operator.
10. Variables in Python must be declared before use.

Section C - Multiple Choice Questions (MCQ)

(10 × 1 = 10 Marks)

1. Which of the following is a valid variable name?
 - a) 2num
 - b) num_2
 - c) num-2
 - d) @num
2. Which loop is used when the number of iterations is known?
 - a) while
 - b) do-while
 - c) for
 - d) switch
3. Which function converts a string to integer?
 - a) float()
 - b) str()
 - c) int()

- d) bool()
- 4. What will type([10,20]) return?
 - a) tuple
 - b) list
 - c) dict
 - d) set
- 5. Which operator is used for logical OR?
 - a) &&
 - b) ||
 - c) or
 - d) |
- 6. What does pop() do?
 - a) Adds element
 - b) Removes element
 - c) Sorts list
 - d) Clears list
- 7. Which symbol is used for comments?
 - a) //
 - b)
 - c) #
 - d) **
- 8. Which keyword is used to handle exceptions?
 - a) error
 - b) handle
 - c) try
 - d) catch
- 9. What will len("Python") return?
 - a) 5
 - b) 6
 - c) 7
 - d) Error
- 10. Which data structure stores key-value pairs?
 - a) List
 - b) Tuple

- c) Dictionary
- d) Set

Section D – Output Based Questions

(10 × 1 = 10 Marks)

a. `for i in range(3):`
 `print(i)`

b. `x = 5`
 `if x > 3:`
 `print("Hello")`
 `else:`
 `print("Bye")`

c. `lst = [10,20,30]`
 `print(lst[1])`

d. `print(10//3)`

e. `s = "Python"`
 `print(s[1:4])`

f. `t = (1,2,3)`
 `print(len(t))`

g. `d = {1:'A',2:'B'}`
 `print(d[2])`

h. `i = 1`
 `while i<=3:`
 `print(i)`
 `i+=1`

i. `a = 5`
 `b = 10`
 `print(a>b)`

j. `print("Hello"*2)`

Section E - Long Programming Questions

(6 × 5 = 30 Marks)

Attempt ALL questions

1. List Based

Write a Python program to create a list of 5 numbers and:

- Display the list
- Find the largest number
- Display the sum of elements

2. Dictionary Based

Write a program to create a dictionary of 5 students with marks.

Display:

- All students
- Student with highest marks

3. Tuple Based

Write a Python program to:

- Create a tuple of 6 elements
- Display only even numbers from the tuple

4. String Based

Write a program to accept a string and:

- Count vowels
- Reverse the string

5. Exception Handling

Write a program to divide two numbers using try and except block to handle division by zero error.

6. User Defined Function

Write a function factorial(n) to calculate factorial of a number.

Call the function and display the result.